

A complete edition from two complementary scans

Source assessment, editorial treatment, and checked Scribus output - 13 September 2026

The two scans support a 32-page content reconstruction: preliminary matter, printed pages 7-21, photographic Plates A-C, and drawing Plates I-X. The clearer primary scan supplies most content; the second fills four gaps.

Resource	Observed content	Use in v01
Primary PDF / 70.0 MB	28 pages; mostly 400 dpi images; Plate I about 600 dpi	Main text, photographs, eight drawings
Secondary PDF / 1.91 MB	34 pages; text mostly 300 dpi; artwork about 150 dpi	Document notice, page 3, Plates II and V
OCR + JP2 derivatives	Separate derivative sets exist for both source PDFs	OCR starts transcription; source PDF pixels preferred

Editable reconstruction. 836 native text frames and 97 editable rules cover the wording, tables, contents, and photo captions. Drawings and their labels retain source pixels. C059 approximates the original typography; 6 x 9 inch text pages and landscape plate pages are modern layout choices.

Checks. The native document was reopened in Scribus 1.6.1 with zero text overflows. Every editable frame string matches the text extracted from its intended PDF page. All pages were rendered and inspected; every plate is present. The PDF has 32 navigation bookmarks.

Remaining limit. Plates II and V retain faint reverse-side printing and paper texture. Their 150 dpi JPEG source limits detail. No replacement geometry, automatic tracing, or guessed lettering has been introduced. Apparent original wording errors are recorded in the project notes.

Sources: [Internet Archive item and scan description](#) | [Complete file listing](#). Exact filenames, hashes, and page mappings are bundled in the project. The two final library/barcode and blank-back-cover pages of the secondary scan are omitted.

Printed page 7: type and specification table

The wording is reset as native text; the table uses editable labels, values, and leaders.

SOURCE SCAN

Q. F. HOTCHKISS 2.244-INCH, 6-POUNDER, 6-HUNDRED-WEIGHT, MARK II GUN.

1. TABLE OF WEIGHTS, DIMENSIONS, ETC.

Weight of gun, including breech mechanism.....	pounds	644
Caliber	inches	2.244
Total length	do	60
Length of bore, including chamber.....	do	52.12
Length of bore, including chamber, in calibers.....		23
Rifling, uniform, 1 turn in 30 calibers:		
Number of grooves.....		24
Width of grooves.....	inch	.22
Depth of grooves.....	do	.012

2. DESCRIPTION OF THE GUN BODY.

(See Plate I.)

(a) The gun is of British design and manufacture. The body is of alloy steel, built up, and consists of a tube, jacket, recoil band, and locking ring. The jacket is shrunk on and envelops the rear end of the tube, being extended rearwards to allow a recess for the breechblock to be formed in it.

(b) The recoil band is screwed on the exterior of the jacket just in front of the breech recess. It is provided with top and bottom lugs for attaching to the recoil and counter-recoil mechanisms, respectively. It is secured against turning by two flush set screws, one at each end of the horizontal diameter.

(c) The jacket and tube are secured longitudinally by the front end of the jacket seating against a shoulder on the tube and the locking ring being screwed and shrunk on both the tube and jacket over this shoulder seat, thus locking these parts together.

(d) Two feathers are formed on the jacket, one on the top, the other on the bottom, which slide in featherways cut into the cradle (Pl. VI) and serve as guides for the gun when in its mounting.

(e) Horizontal axis lines are cut on the sides of the gun at the muzzle end. Vertical and horizontal lines are also cut on the breech and muzzle faces.

(f) A line locating the center of gravity (without mechanism) is cut transversely on the upper side of the gun.

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SCRIBUS RECONSTRUCTION

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Source line endings are retained; ink marks no longer distort the reconstructed text baselines.

SOURCE SCAN

- (b) *To fill the recoil cylinder.*—Fully depress the gun, remove the filling plug from the cylinder, and pour in oil until the cylinder is full. Replace filling-hole plug. Quantity of cylinder oil required is 1 pint.
- (c) *To empty the recoil cylinder.*—Depress the gun. Unscrew and remove front closing plug and run the oil off into a suitable receptacle.
- (d) *Leakage of gland.*—Should it be found that there is leakage at the recoil cylinder gland the latter should be tightened, but if this does not stop the leak the packing must be renewed. It must, however, be remembered that the gland should not be too tight or the gun may fail to return correctly.
- (e) If it is found that leakage is taking place over the leather washer of front plug, or stuffing box, the fitting must be removed and a new washer placed in position.
- (f) *Counter-recoil springs.*—The counter-recoil rods must be firmly nutted up to the gun lug. The springs must be frequently examined and any spring which is found broken or with a permanent set must be exchanged.
- (g) The whole of the working parts must be kept thoroughly clean and well lubricated.

List of lubricators on mounting.

Fitting.	Number of lubricators.	Position.
Cradle.....	1	Right rear end.
Do.....	1	Top front end.
Do.....	1	Oil screw, left rear end.
Trunnion bearings.....	1	Oil screw, right hand.
Do.....	1	Oil screw, left hand.
Firing gear.....	2	Lever bearings.
Elevating and traversing clamps.....	2	1 on each clamp.
Pivot.....	1	Through center of pivot bolt.

13. INSTRUCTIONS FOR ASSEMBLING THE GUN AND MOUNTING.

- (a) Place clamp screw brushes in revolving bracket and secure with grub screws. Place revolving bracket on base plate.
- (b) Place clip ring in position, see that keys on bracket take key-ways in ring, then secure with two bolts.
- (c) Screw in adjustable pivot and adjust so that revolving bracket revolves freely, then secure by keeper screw.
- (d) Assemble cradle. Screw stuffing box into rear end of recoil cylinder, see that leather washer is in place, pass piston rod through cylinder and stuffing box. Pack stuffing box and set up gland. Screw plug into front end of cylinder. See that leather washer is in place.

SCRIBUS RECONSTRUCTION

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- (e) If it is found that leakage is taking place over the leather washer of front plug, or stuffing box, the fitting must be removed and a new washer placed in position.
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Firing gear.....	2	Lever bearings.
Elevating and traversing clamps.....	2	1 on each clamp.
Pivot.....	1	Through center of pivot bolt.

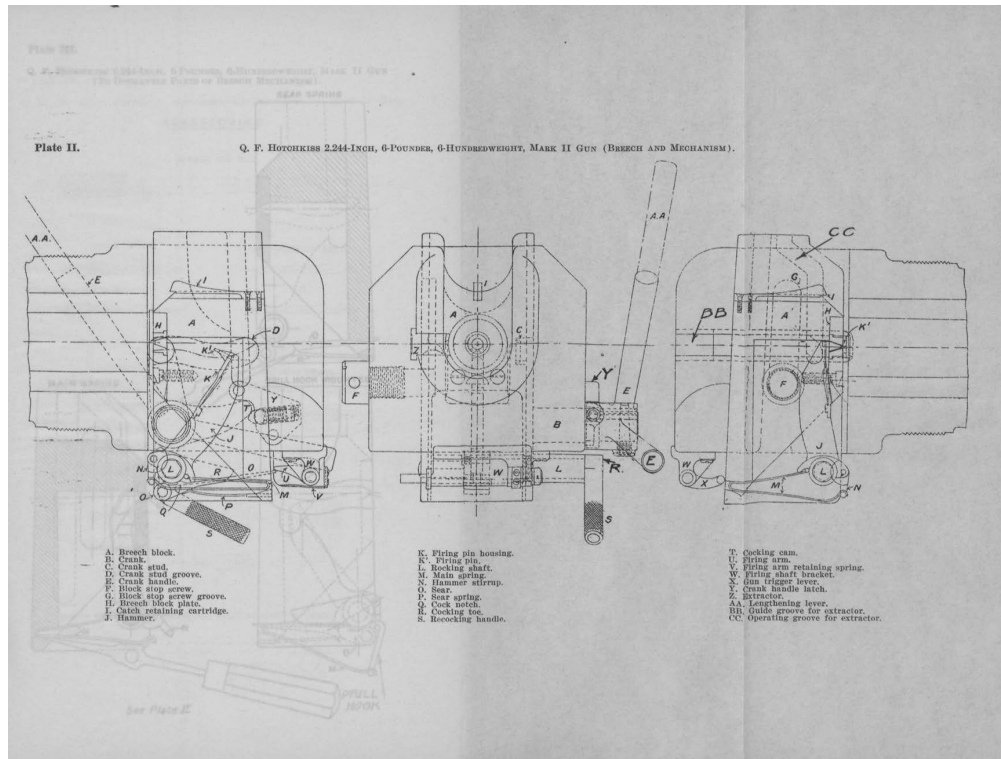
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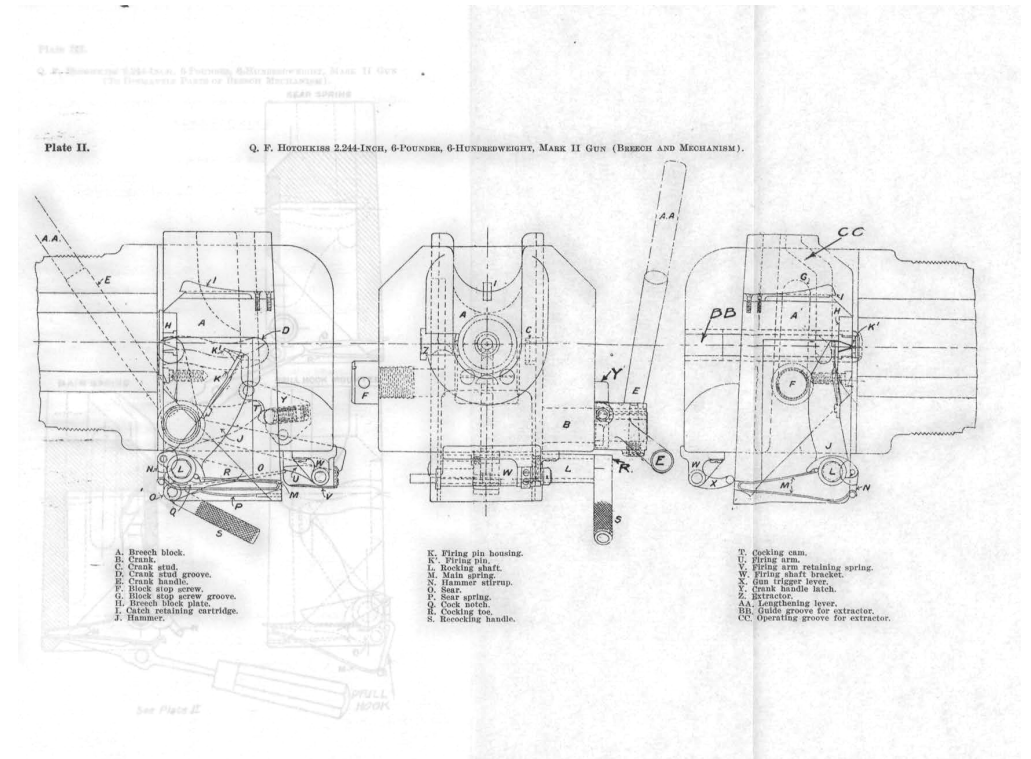
Plate II: conservative cleanup, visible limitations

The second scan supplies this otherwise missing plate. Faint reverse-side printing is preserved where separation is uncertain.

SECONDARY SOURCE



CLEANED CANDIDATE IN THE MASTER



The treatment normalizes uneven paper tone and modestly increases contrast. All front-side marks remain sampled image content; no hidden lines or lettering have been reconstructed. Plate V has the same source limitation and receives the same treatment.

A sharper scan of these two leaves would offer more improvement than additional thresholding. The package retains the untouched source images and deterministic processing script so that future comparisons or replacements remain traceable.